Venkata Lakshmi Parimala Pasupuleti

Senior AI Engineer · Agentic AI Systems · Python / LangGraph / Java · Supply Chain & Enterprise Automation

venkatalakshmiparimala@gmail.com | +1-703-261-5410 | Virginia, USA | LinkedIn | AgentProbe (GitHub) | GitHub

SUMMARY

Senior AI Engineer with 9 years of full-stack backend experience and hands-on production agentic AI delivery. Built LeaseI - an 8-agent LangGraph support investigation platform with HITL escalation, RAG retrieval, and a CI-integrated eval harness across 1,240 test cases - the same architecture FourKites deploys for customer support automation. Built AgentProbe, an open-source LLM evaluation toolkit used to benchmark agent quality across correctness, hallucination rate, groundedness, latency, and token efficiency. At Entrata, shipped 20+ full-stack modules across 120+ Agile sprints in Python, Java, PostgreSQL, and JavaScript - owning backend applications end-to-end through development, QA, and release. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

AI & Agents: LangGraph, Agentic AI, AI Agents, Multi-Agent Orchestration, Agentic Frameworks, Agentic Workflows, LLM Orchestration, Tool Calling, HITL, Human-in-the-Loop, Confidence Scoring, Guardrails, RAG Pipelines, ChromaDB, Vector Database, Semantic Search, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Prompt Engineering, Few-Shot Prompting, Output Validation, LLMOps, Agent Observability, Agent Tracing, OpenAI API, Anthropic API, Groq, Ollama, MCP

Languages & Frameworks: Python, Java, JavaScript, TypeScript, SQL, PL/SQL, PHP, R, Bash, HTML/CSS, React.js, FastAPI, RESTful APIs, Angular, Spring Boot, .NET Framework, Object-Oriented Programming (OOP), jQuery, AJAX

Data & ML: PostgreSQL, MySQL, Microsoft SQL Server, Oracle, NoSQL, ETL, Data Modeling, Data Pipelines, PyTorch, TensorFlow, scikit-learn, XGBoost, NumPy, Pandas, spaCy, BERT, Hugging Face, Tableau, Power BI

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Google Cloud Storage, Docker, Kubernetes (K8s), CI/CD (GitHub Actions), RabbitMQ, Redis, Microservices, Distributed Systems, Observability, Production Monitoring, Full Agile SDLC, Apache Spark, Libsodium, Git, Linux, JIRA, Confluence

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Built a production agentic workflow using LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) - risk-aware compliance strategy with compliant missions routed to export and non-compliant ones rejected with grounded, cited reasons.
- Implemented a RAG layer over regulatory corpus (FAA Part 107) answering operator queries with cited sources; engineered validation guardrails enforcing altitude limits, battery budgets, and geometric no-fly-zone checks before export.

Entrata (PropTech / Real Estate SaaS · 10M+ units managed)
Sep 2018 - Aug 2024

Senior Software Engineer

- Shipped 20+ full-stack production modules (Python, PHP, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints - owned backend applications end-to-end through development, QA, and release; led microservices redesign using Docker (50% scalability improvement, 30% deployment time reduction).
- Built and owned the RabbitMQ-based async processing system for gross-rent updates across thousands of tenant records - designed retry logic and idempotency guarantees that eliminated double-processing failures and kept the system stable through production incidents and annual rent-schedule changes.
- Optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, partitioning, and stored procedure refactoring; designed and maintained database schemas for eligibility, financial, and document workflows across multi-tenant production environments.
- Led the AWS S3 document migration - mapped data models, validated migrated content and metadata, confirmed customer-facing workflows post-cutover; achieved 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents).
- Mentored 5+ junior engineers; conducted 300+ peer code reviews; built internal debugging tools that reduced debugging time by 40%; eliminated 80% of manual operational workload through Python/Bash automation pipelines.

Entrata India (Db Xento Systems)
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee Software Engineer

- Built property management modules in PHP, Python, Java, and PostgreSQL - including income-limits import (1.6M HUD/Tax Credit records across 5 years), document management workflows, and compliance pipelines.
- Led cloud migration of client documents from legacy server storage to AWS S3; redesigned document storage database architecture, migrating hundreds of thousands of records to a modern schema.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets (190+ countries) - $R^2 = 0.977$ regression accuracy; applied Isolation Forest for automated data integrity checks across 3 research pipelines.
- Delivered Python programming and database design workshops to 600+ students across multiple sections of MIS 303.

PROJECTS

Leasel: Agentic Support Investigation Platform

2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · SQLite · OCR · Streamlit

- Built a production 8-agent LangGraph StateGraph for support automation: Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation - with conditional routing, HITL escalation on low-confidence outputs, and evidence-backed root-cause hypotheses. Directly mirrors the support automation agent architecture described in FourKites' role.
- RAG layer (TF-IDF retrieval, section-aware chunking, OCR fallback) over compliance documents - 100% citation correctness, zero unsupported claims; core rule: LLM reasons, deterministic logic makes every final regulated call.
- Evaluated across 1,240 residential lease PDFs: 93.6% precision, 87.9% F1, pass@1 81.7%, pass@3 88.3% via multi-trial sampling with LLM-as-judge scoring and CI-integrated regression gate.

AgentProbe: Open-Source LLM Evaluation Toolkit

2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions)

- Built a fully local, zero-cost AI agent evaluation toolkit: 15-endpoint FastAPI backend, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate, multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), LLM-as-judge scoring with multi-trial pass@k harness - used to benchmark Leasel across 1,240+ test cases.

AI on FHIR: NLP-Driven Medical Records Query System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR · spaCy · BERT · Docker

- Reduced manual clinical retrieval time ~60% by building a Flask + React full-stack platform translating natural-language queries into FHIR API calls using LLMs, BERT, and embeddings; HIPAA-compliant RBAC across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017